

Status	Finished
Started	Sunday, 12 January 2025, 1:47 PM
Completed	Sunday, 12 January 2025, 2:07 PM
Duration	20 mins 8 secs

Question **1**

Incorrect

Marked out of 1.00

 Flag question

Given an array of numbers, find the index of the smallest array element (the pivot), for which the sums of all elements to the left and to the right are equal. The array may not be reordered.

Example

arr=[1,2,3,4,6]

- the sum of the first three elements, $1+2+3=6$. The value of the last element is 6.
- Using zero based indexing, arr[3]=4 is the pivot between the two subarrays.
- The index of the pivot is 3.

Function Description

Complete the function balancedSum in the editor below.

balancedSum has the following parameter(s):

int arr[n]: an array of integers

Returns:

int: an integer representing the index of the pivot

Constraints

- $3 \leq n \leq 10^5$
- $1 \leq arr[i] \leq 2 \times 10^4$, where $0 \leq i < n$
- It is guaranteed that a solution always exists.

Input Format for Custom Testing

Input from stdin will be processed as follows and passed to the function.

The first line contains an integer n , the size of the array `arr`.

Each of the next n lines contains an integer, `arr[i]`, where $0 \leq i < n$.

Sample Case 0

Sample Input 0

STDIN Function Parameters

4 → `arr[]` size $n = 4$

1 → `arr = [1, 2, 3, 3]`

2

3

3

Sample Output 0

2

Explanation 0

- The sum of the first two elements, $1+2=3$. The value of the last element is 3.
- Using zero based indexing, `arr[2]=3` is the pivot between the two subarrays.
- The index of the pivot is 2.

Sample Case 1

Sample Input 1

STDIN Function Parameters

3 → `arr[]` size $n = 3$

1 → `arr = [1, 2, 1]`

2

Sample Output 0

2

Explanation 0

- The sum of the first two elements, $1+2=3$. The value of the last element is 3.
- Using zero based indexing, $\text{arr}[2]=3$ is the pivot between the two subarrays.
- The index of the pivot is 2.

Sample Case 1

Sample Input 1

STDIN Function Parameters

3 → arr[] size n = 3

1 → arr = [1, 2, 1]

2

1

Sample Output 1

1

Explanation 1

- The first and last elements are equal to 1.
- Using zero based indexing, $\text{arr}[1]=2$ is the pivot between the two subarrays.
- The index of the pivot is 1.

Answer: (penalty regime: 0 %)

Reset answer

```

1  /*
2  * Complete the 'balancedSum' function below.
3  *
4  * The function is expected to return an INTEGER.
5  * The function accepts INTEGER_ARRAY arr as parameter.
6  */
7
8  int balancedSum(int arr_count, int* arr)
9  {
10     int totalsum = 0;
11     for(int i = 0; i < arr_count; i++){
12         totalsum += arr[i];
13     }
14     int leftsum = 0;
15     for(int i=0; i<arr_count; i++){
16         int rightsum = totalsum - leftsum - arr[i];
17         if(leftsum==rightsum){
18             return i;
19         }
20         leftsum += arr[i];
21     }
22     return 1;
23 }
24

```

	Test	Expected	Got	
✓	int arr[] = {1,2,3,3}; printf("%d", balancedSum(4, arr))	2	2	✓

Passed all tests! ✓